Hideaki Takahashi

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Summary

- Second-year Ph.D. student at Columbia University researching the intersection of **AI**, **systems security**, and **cryptography**, such as bug detection and formal verification of **Zero-Knowledge Proof** and **Agentic System**.
- Multiple first-authored papers in top-tier conferences (**IEEE S&P**, **AAMAS**, **CVPR**).
- Discovered **70+ confirmed zero-day vulnerabilities and CVEs** in widely used Web3 and AI Agent projects.
- Skilled open-source developer of widely adopted tools (**1k+ GitHub stars**).
- Awarded **five medals in Kaggle**, the world's largest platform for machine learning competitions.

Education

Columbia University in the City of New York — *New York, United States*

Ph.D (Computer Science) — *Sep. 2024 - Present*

- Supervised by Prof. Junfeng Yang
- Member of The Center for Digital Finance and Technologies

The University of Tokyo — *Tokyo, Japan*

Bachelor of Arts and Sciences (Informatics) — *Apr. 2019 - Mar. 2024*

- Supervised by Prof. Alex Fukunaga., GPA: 3.85/4.0

Papers

[1] Hideaki Takahashi*, Jihwan Kim, Suman Jana, and Junfeng Yang. zkFuzz: Foundation and Framework for Effective Fuzzing of Zero-Knowledge Circuits . In *2026 IEEE Symposium on Security and Privacy (SP)*, pages 919–938, Los Alamitos, CA, USA, May 2026. IEEE Computer Society. **Peer-reviewed @ IEEE S&P'26** (CORE Rank: A*, Acceptance Rate: 13%).

[2] Hideaki Takahashi* and Alex Fukunaga. On the transit obfuscation problem. In *International Conference on Autonomous Agents and Multi-Agent Systems*, 2024. **Peer-reviewed @ AAMAS'24** (CORE Rank: A*, Acceptance Rate: 20.7%).

[3] Tianyuan Zou, Zixuan Gu, Yu He, Hideaki Takahashi, Yang Liu, Guangnan Ye, and Ya-Qin Zhang. VFLAIR: A research library and benchmark for vertical federated learning. In *The Twelfth International Conference on Learning Representations*, 2024. **Peer-reviewed @ ICLR'24** (CORE Rank: A*, Acceptance Rate: 31.1%).

[4] Hideaki Takahashi*, JingJing Liu, and Yang Liu. Breaching fedMD: Image recovery via paired-logits inversion attack. In *Conference on Computer Vision and Pattern Recognition*, 2023. **Peer-reviewed @ CVPR'23** (CORE Rank: A*, Acceptance Rate: 25.8%).

[5] Sally Junsong Wang, Jianan Yao, Kexin Pei, Hideaki Takahashi, and Junfeng Yang. Detecting buggy contracts via smart testing. *arXiv preprint arXiv:2409.04597*, 2024.

[6] Hideaki Takahashi*. Aijack: Security and privacy risk simulator for machine learning. *arXiv preprint arXiv:2312.17667*, 2023.

[7] Hideaki Takahashi*, JingJing Liu, and Yang Liu. Eliminating label leakage in tree-based vertical federated learning. *arXiv preprint arXiv:2307.10318*, 2023.

[8] Hideaki Takahashi*, Kohei Ichikawa, and Keichi Takahashi. Difficulty of detecting overstated dataset size in federated learning. Technical Report 10, 2021. `http://id.nii.ac.jp/1001/00214220/`.

Talks

UC Berkeley Security Seminar — *United States*

Invited Speaker — *Mar. 11 2026*

- Title: zkFuzz: Foundation and Framework for Effective Fuzzing of Zero-Knowledge Circuits

Zero Knowledge Frontier Japan: vol.2 *Japan*
INVITED SPEAKER *Nov. 29 2025*
- Title: Typical Bugs and Detection Methods in Implementing Zero-Knowledge Proof Constraints

Research Experience

Columbia University *New York, United States*
PH.D STUDENT *Sep. 2024 - Present*
- Conducted research on the software testing [1, 5] under the supervision of Prof. Junfeng Yang.

Fukunaga Lab, The University of Tokyo *Tokyo, Japan*
UNDERGRADUATE STUDENT *Apr. 2023 - Mar. 2024*
- Conducted research on the transit obfuscation problem [2], a new task of privacy-preserving AI planning, under the supervision of Prof. Alex Fukunaga.

Institute for AI Industry Research, Tsinghua University *Beijing, China*
FEDERATED LEARNING & PRIVACY COMPUTING INTERNS *Jan. 2022 - Feb. 2023*
- Conducted research on federated learning and privacy computing [3, 4] under the supervision of Prof. Yang Liu and Prof. Jingjing Liu.

Nara Institute of Science and Technology *Nara, Japan*
VISITING STUDENT *Aug. 2021 - Sep. 2021*
- Conducted research on the free-rider problem of federated learning [8] under the supervision of Prof. Kohei Ichikawa and Prof. Keichi Takahashi.

Industry Experience

Apple Inc. *Yokohama, Japan*
TECHNICAL INTERNSHIP: AIML/SOFTWARE ENGINEER *Feb. 2024 - Jul. 2024*
- Worked on AIML/software engineering.

UTokyo Economic Consulting Inc. *Tokyo, Japan*
RESEARCH ASSISTANT *Oct. 2020 - Mar. 2024*
- Worked on social implementations of econometrics and machine learning.

Awards & Fundings

FUNDINGS

2025	**The Columbia-Ethereum Research Center for Blockchain Protocol Design**, PhD fellowship
2024 - 2026	**Funai Overseas Scholarship**, Granted 2 years of tuition and stipend.

COMPETITIONS

2023	**45th / 616 teams (Silver Medal)**,	Kaggle: Google - Fast or Slow? Predict AI Model Runtime
2021	**67th / 875 teams (Bronze Medal)**,	Kaggle: Hungry Geese
2021	**52nd / 788 teams (Bronze Medal)**,	Kaggle: Santa 2020 - The Candy Cane Contest
2020	**51st / 1138 teams (Silver Medal)**,	Kaggle: Google Research Football with Manchester City F.C.
2020	**88th / 1390 teams (Bronze Medal)**,	Kaggle: Cornell Birdcall Identification

Software

AIJack (`https://github.com/Koukyosyumei/AIJack`)
OWNER
- Security risk simulator for machine learning (400+ stars on GitHub, 10K+ downloads, referenced in 8+ papers)

h5i (`https://github.com/h5i-dev/h5i`)
OWNER
- Next-Gen AI-Aware Git. Model/prompt-aware commits, 95% less token waste, 3.5x richer PR brief, 1.8x faster multi-agent loops. (300+ stars on GitHub)